Virtual Audio Router

Developer Documentation

Version 2.1.0 — Internal Reference

1. Project Structure

```
D:\wdm2vst\
├── ASMRTOP_Driver_V2/          # Windows Kernel Driver (WDM/WaveRT)
│   ├── Source/
│   │   ├── Main/               # Driver entry, miniport, IPC
│   │   ├── Inc/                # Headers, branding.h
│   │   ├── Filters/            # Audio filter implementations
│   │   └── Utilities/          # Helpers
│   ├── VirtualAudioDriver.sln  # VS2022 solution
│   ├── build.bat               # Build script
│   └── set_brand.ps1           # Brand switcher
│
├── ASMRTOP_Plugins/            # VST3 Plugins (JUCE C++)
│   ├── Source/
│   │   ├── W2V_Plugin*.cpp/h   # WDM2VST
│   │   ├── V2W_Plugin*.cpp/h   # VST2WDM
│   │   ├── Send_Plugin*.cpp/h  # Audio Send
│   │   ├── Recv_Plugin*.cpp/h  # Audio Receive
│   │   ├── AsmrtopIPC.h        # Shared memory IPC
│   │   ├── P2PTransport.h      # UDP + WebSocket
│   │   ├── Localization.h      # CN/EN strings
│   │   ├── PluginBranding.h    # Brand macros
│   │   ├── QRCode.h            # QR generator
│   │   └── QRCodePopup.h       # QR dialog
│   ├── JUCE/                   # Framework (submodule)
│   └── CMakeLists.txt          # Build config
```

2. Architecture

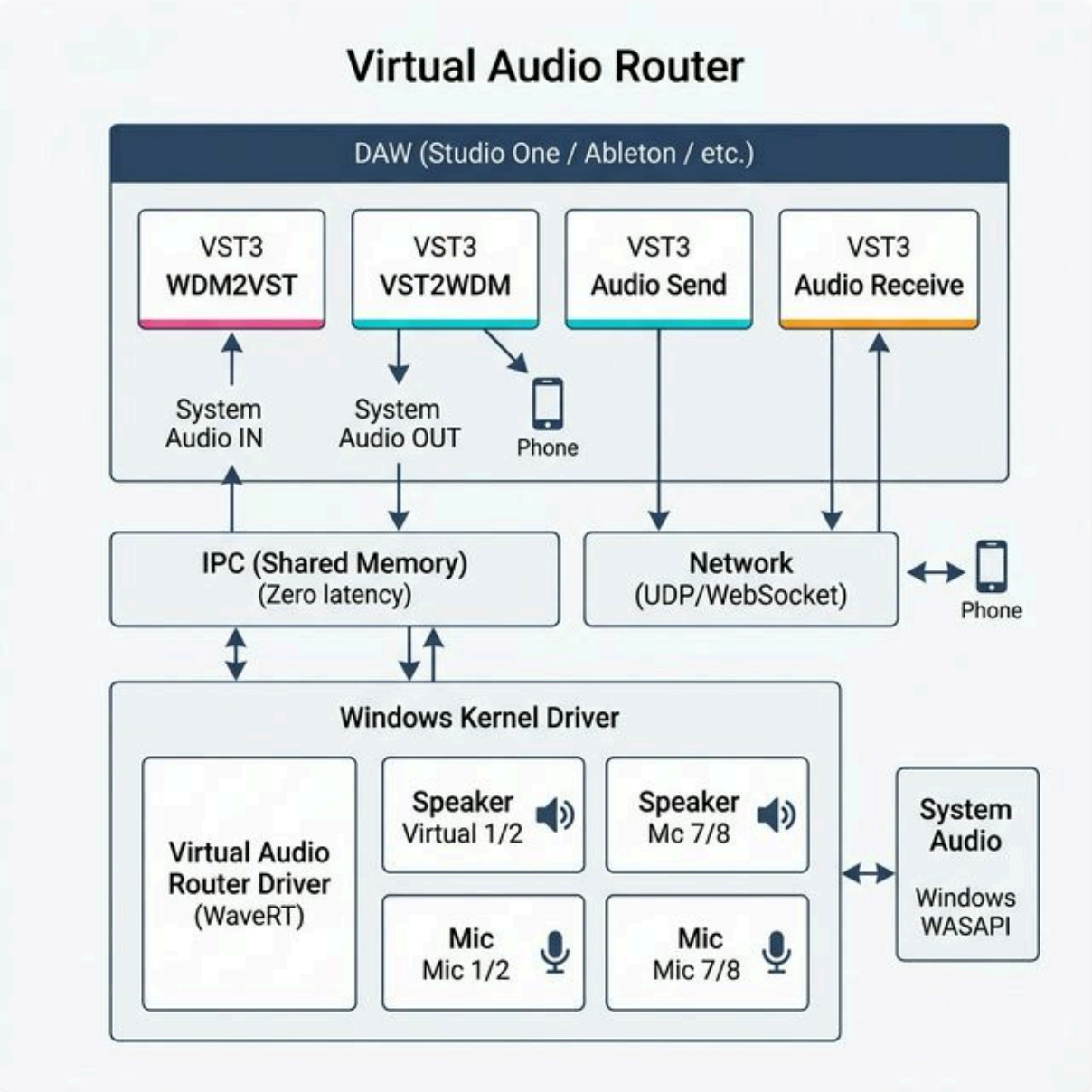

Figure 1: System Architecture

2.1 IPC Data Flow (Zero Latency)

```
DAW processBlock() → SharedMemoryBridge::write() → Kernel Driver reads
Kernel Driver writes → SharedMemoryBridge::read() → DAW processBlock()
```

- Named shared memory: `Global\VirtualAudioWDM_PLAY_0`

- Lock-free ring buffer with atomic read/write positions
- Packet: 32-byte header + interleaved float32 PCM

2.2 Network Data Flow (UDP)

```
Send::processBlock() → P2PSender → DatagramSocket → UDP packet
UDP packet → DatagramSocket → P2PReceiver → Recv::processBlock()
```

- 16-bit PCM encoding (bandwidth efficient)
- Default port: 18810

2.3 Mobile Data Flow (WebSocket)

```
processBlock() → pushToWebSocketClients() → TCP WebSocket frame
Phone Browser ← WebSocket ← Built-in HTTP/WS server
```

- Hand-rolled HTTP server + WebSocket upgrade
- SHA-1 inline for Sec-WebSocket-Accept
- JavaScript AudioWorklet playback in browser

3. Core Components

3.1 SharedMemoryBridge (AsmrtopIPC.h)

```
struct SharedAudioPacket {
    uint32_t magic;       // 0x41534D52 ("ASMR")
    uint32_t version, channelId, sampleRate;
    uint32_t numSamples, numChannels;
    uint32_t writePos, readPos;
    float data[RING_BUFFER_SAMPLES * 2];
};

class SharedMemoryBridge {
    SharedMemoryBridge(baseName, direction, channelId);
    void write(left, right, numSamples);
    int read(left, right, maxSamples);
};
```

3.2 P2PTransport (P2PTransport.h, ~660 lines)

P2PSender manages: IPC + UDP + WebSocket
P2PReceiver manages: IPC + UDP listener

3.3 Localization (Localization.h)

```
#define ZH(s) juce::String::fromUTF8(s)

class Lang {
    enum ID { CHANNEL, MODE, STATUS, ... };
    static juce::String get(ID id);
    static void toggle();  // CN ⇔ EN
};
```

Uses `fromUTF8()` with raw hex UTF-8 sequences. Brand-aware titles via `PluginBranding.h` macros.

3.4 Branding (PluginBranding.h + set_brand.ps1)

Macro	Public	Branded
PLUGIN_COMPANY_NAME	"VirtualAudioRouter"	"ASMRTOP"
PLUGIN_IPC_BASE_NAME	"VirtualAudioWDM"	"AsmrtopWDM"
VST3 file names	WDM2VST.vst3 etc.	ASMRTOP WDM2VST.vst3

Critical: Plugin IPC names must match driver IPC names. Never mix brands!

4. Building

4.1 Prerequisites

Tool	Version
Visual Studio	2022 (v143)
CMake	3.20+
Windows SDK	10.0.22621.0+
WDK	Driver builds only

4.2 Build Commands

```
# Switch brand
powershell -EP Bypass -File set_brand.ps1 public

# Configure + Build
cd ASMRTOP_Plugins
cmake -B build -S .
cmake --build build --config Release
```

4.3 Output

```
build/<TARGET>_artefacts/Release/VST3/<Name>.vst3/
```

5. Plugin Details

5.1 WDM2VST — processBlock

1. IPC mode → read from SharedMemoryBridge
2. WASAPI mode → read from ring buffer (filled by audioDeviceIOCallback)
3. Apply gain + limiter
4. Update peak meters

5.2 VST2WDM — processBlock

1. Apply gain + limiter
2. Write to WASAPI ring buffer
3. Write to IPC SharedMemoryBridge (if IPC mode)
4. Push to WebSocket clients (if mobile enabled)
5. Update peak meters

5.3 Audio Send

Manages `P2PSender` lifecycle. processBlock pushes audio to all enabled transports.

5.4 Audio Receive

Manages `P2PReceiver` lifecycle. processBlock reads from IPC/UDP ring buffer.

6. Design Decisions

1. **Header-only libs** — IPC, P2P, QR, i18n for easy CMake integration
2. **No external network deps** — WebSocket/HTTP hand-rolled with JUCE sockets
3. **UTF-8 hex escapes** — Avoids source encoding issues across editors
4. **Compile-time branding** — Single codebase, two products via macros
5. **Lock-free ring buffer IPC** — Zero-latency kernel↔userspace audio transfer
6. **16-bit PCM for network** — Half bandwidth vs float32, good enough for monitoring

7. File Reference

File	Lines	Purpose
AsmrtopIPC.h	~120	Shared memory ring buffer bridge
P2PTransport.h	~660	UDP sender/receiver + WebSocket server
Localization.h	~155	CN/EN string table
PluginBranding.h	~30	Brand macro definitions
QRCode.h	~300	Minimal QR code generator
QRCodePopup.h	~60	QR popup JUCE component
set_brand.ps1	~160	PowerShell brand switcher